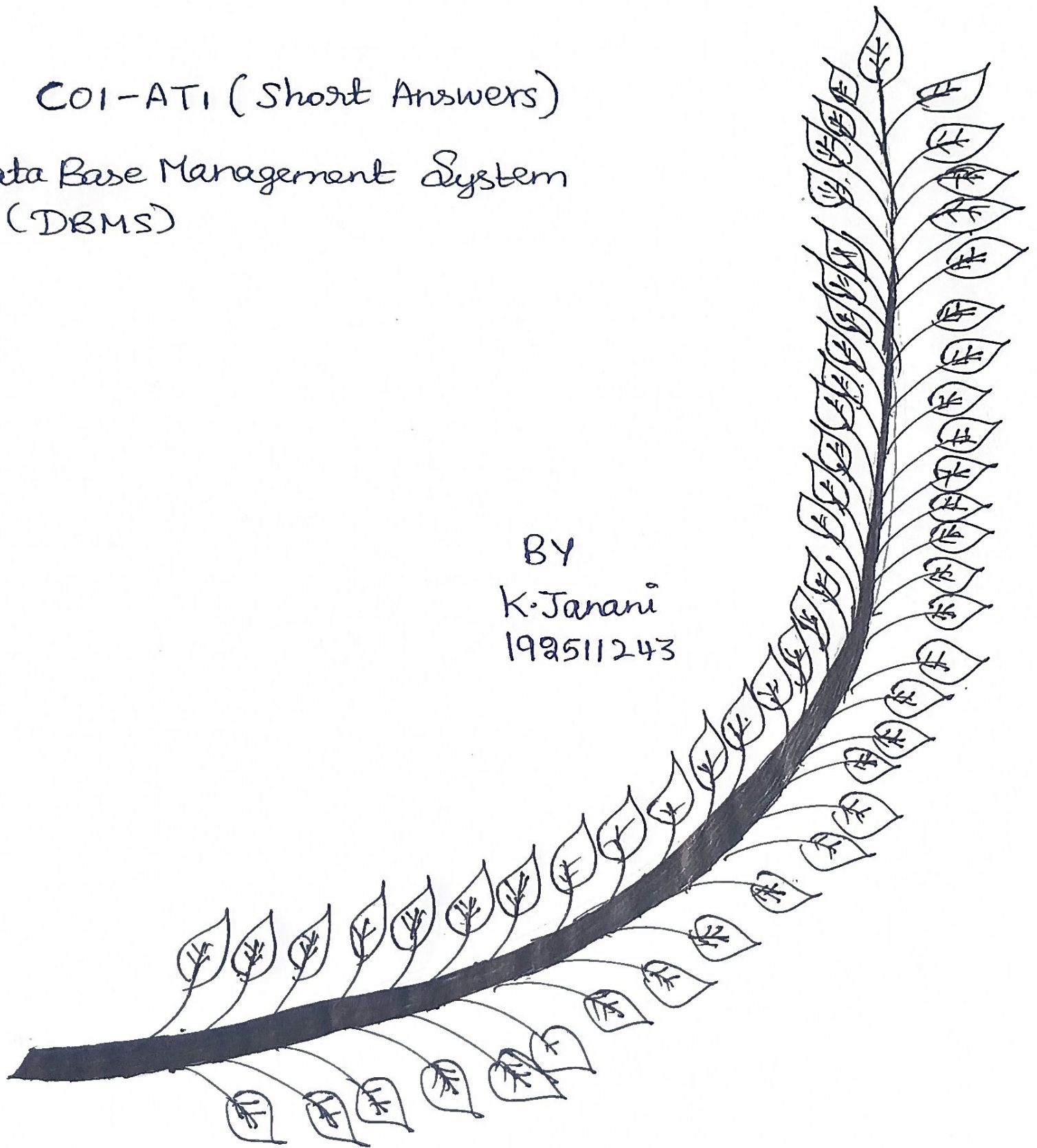


COI-AT1 (Short Answers)
Data Base Management System
(DBMS)

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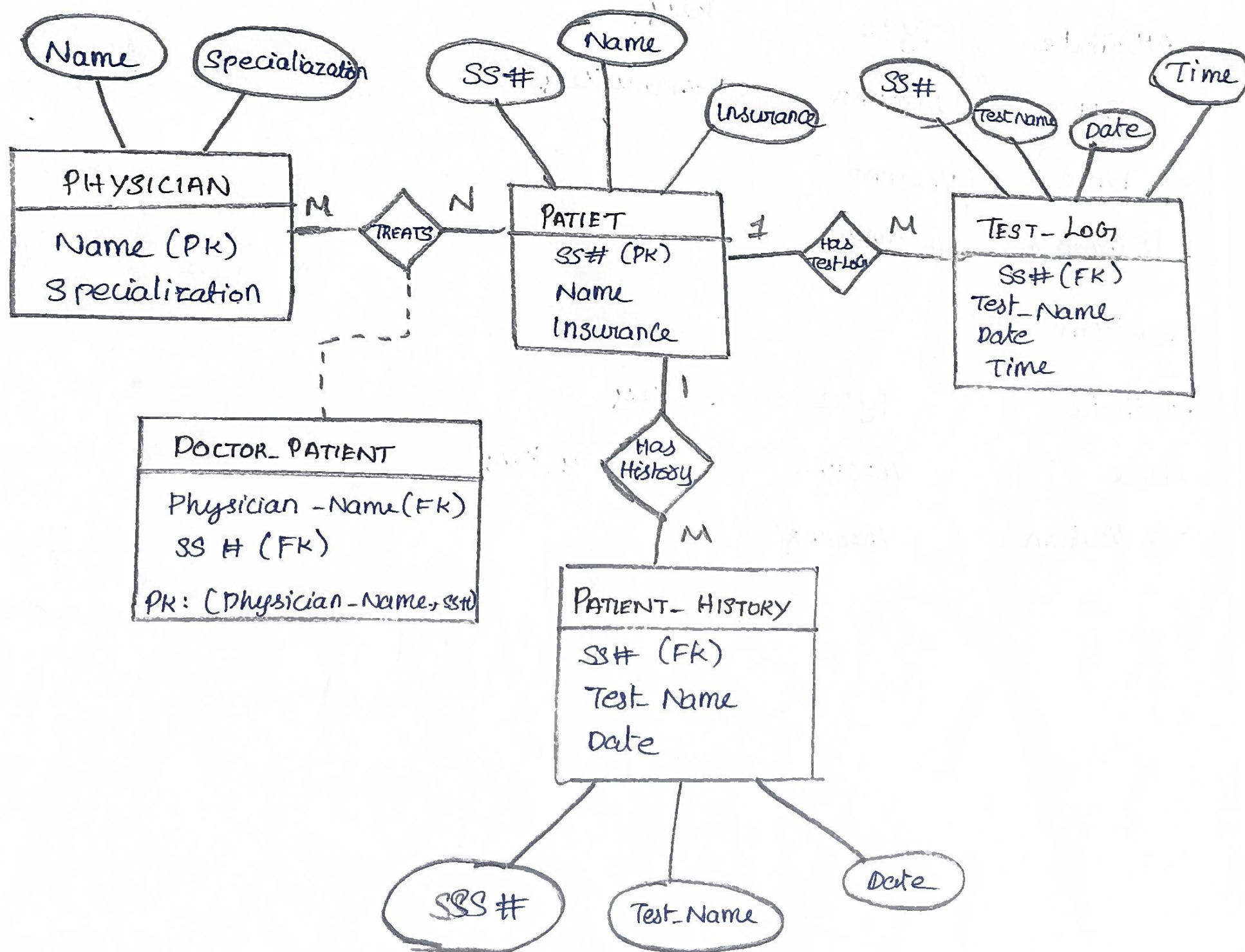


Hospital Management System - ER Diagram and database Tables

1. Problem Statement

Design an Entity - Relationship (ER) Diagram for a Hospital management System containing patients, Physicians, tests, and patient history. Also construct the corresponding relational Tables.

2. ER Diagram (Text Representation).



3. Relationship Description

Entity	Relationship
Patient - Test log	one patient Can have many test records
Patient - patient History	one patient Can have many History records.
Physician - Patient	one Physician Treats many patients and one Patient maybe treated by many physicians.

4. Relational Tables.

Patient

Attribute	Type	Key.
SSN	VARCHAR	Primary Key.
Name	VARCHAR	
Insurance	VARCHAR	

Physician

Attribute	Type	Key
Name	VARCHAR	Primary Key
Specialization	VARCHAR	

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 physicians.

st. Log

Attribute	Type	key
SS #	VARCHAR	Foreign key
Test-Name	VARCHAR	
Date	DATE	
Time	TIME	

Foreign Key: SS# → Patient (SS#)

Doctor - Patient

Attribute	Type	key
Physician-Name	VARCHAR	Foreign key
SS #	VARCHAR	Foreign key.

Foreign Keys:

* Physician-Name → Physician (Name)

* SS # → Patient (SS#)

Composite Primary key : (Physician-Name, SS#)

Patient - History :

Attribute	Type	key
SS #	VARCHAR	Foreign key
Test-Name	VARCHAR	
Date	Date	

Foreign Key : SS# → Patient (SS#)

5. Key Constraints :

- * Patient : SSN uniquely identifies each patients
- * physician : Name uniquely identifies each physician
- * Test Log : Stores all tests performed on patients with date and Time
- * Doctor-Patient : Resolves the many to many Relationship Between doctors and Patients.
- * Patient History : Maintains the Medical History of Each Patient.

Conclusion:

The Hospital Management System database is Normalized using five related Tables. The ER model Efficiently Stores patient details, Physicians information, Test records, doctor-patient relationship, and Patients History while Maintaining data integrity through primary and Foreign key.